to be used to access the hardware. And each driver will be assigned a unique major number. All device files carrying the same major number are handled by the same driver. As this number is unique, we earlier used *cat /proc/devices* to see the major number of the active devices.

You may also note that we have a minor number (which is a sort of internal number) that is used by the driver to identify the different hardware it can access. (See the output of *cat /proc/modules* in Figure 1.)

You can also see the minor and major numbers of the devices using *ls -l* command (Figure 2).

During the 'registration' (see the code given above), the *unsigned int* corresponds to the major number and the *const char ** refers to the name of the device (to appear in the output of *cat /proc/devices*). As we discussed earlier, we use the *struct file_operations *fops* to refer to the 'file operations table'.

When you develop an actual device driver, it is advisable that you consult the documentation provided by your distribution to find out an unused major number.

Some of you might be wondering how we are going to use the driver we wrote, without having hardware (dev) to test. We can create a virtual one using the *mknod* command.

Before doing that, let us compile our driver. (If you find it hard to create a simple 'make file' for this, please refer to the previous article.)

If you look at the output (Figure 3), you will notice an error. This is because we forgot to add a licence statement in our code. You can remove the error by adding the following line to the code:

```
MODULE_LICENSE("GPL v3");
```

Now the driver has been compiled. Let's add (load) the driver to the kernel by issuing the following command:

```
sudo insmod LFY_dev.ko
```

It is time to create our device! We can do that by using the following command as the root user:

```
mknod /dev/LFY_device c 161 0
```

This actually creates a new file in the */dev/* directory (Figure 4).

We have also specified (in the command) that it should be a character device (c) and the major number should be 161. You can see the new major and minor numbers of the new device we created, by using the *ls -l* command—see Figure 5.

Our driver and device are ready for use. Let's send some data to the device (after becoming the root):

```
echo "Voyage to Kernel" > /dev/LFY_device
```

If you look at the code carefully you can see that this command can effectively keep on sending the data again and again. And now we can 'read' the device by issuing the following command:

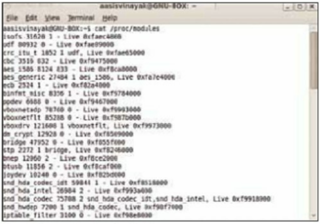

Figure 1: Output of cat /proc/modules/

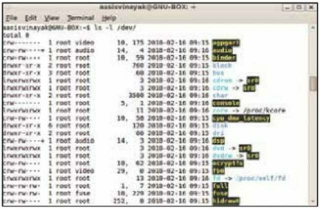

Figure 2: Listing the devices with major and minor numbers

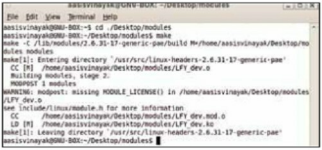

Figure 3: Compiling the driver

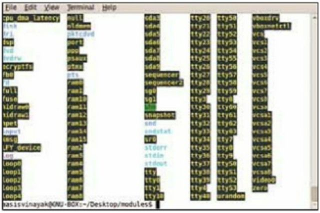

Figure 4: Listing of /dev/ directory